

Lucas Resck

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Education

University of Cambridge, Language Technology Lab

Cambridge, UK

PhD in Computation, Cognition and Language

Oct 2024 – Sept 2028

- Supervisors: Profs. Anna Korhonen and Isabelle Augenstein (University of Copenhagen)
- Affiliations: ELLIS PhD Student, Cambridge Trust Scholar, Girton College Member

University of Copenhagen, CopeNLU

Copenhagen, Denmark

Visiting ELLIS PhD Student in NLP

May 2026 – June 2026

- Supervisor: Prof. Isabelle Augenstein

Getulio Vargas Foundation (FGV), Visual Data Science Lab

Rio de Janeiro, Brazil

MSc in Mathematical Modelling

Feb 2022 – May 2024

- Supervisors: Profs. Jorge Poco and Marcos Raimundo (State University of Campinas, Brazil)

Rice University; Visual, Language, and Learning Lab

Houston, US

Visiting Scholar in Computer Science

Oct 2022 – Dec 2022

- Supervisor: Prof. Vicente Ordóñez

Getulio Vargas Foundation (FGV)

Rio de Janeiro, Brazil

BSc in Applied Mathematics

Dec 2017 – Jan 2022

- Supervisor: Prof. Jorge Poco
- 1st in class, GPA 3.86/4.0 (9.66/10.0, lowest passing grade of 6)

Federal Center for Technological Education of Minas Gerais (CEFET-MG)

Varginha, Brazil

Technician Diploma in Mechatronics

2015 – 2017

- High school researcher fellow and volunteer

Research Projects

Mitigating cross-lingual cultural inconsistencies in LLMs

Oct 2024 – present

- Investigated and formalised the problem of Cross-lingual Cultural Inconsistency (CCI) in LLMs (arXiv).
- Proposed a novel, mathematically robust metric and a self-supervised preference optimisation framework to address this issue, outperforming strong prompting and representation steering baselines.
- Empirically demonstrated the link between CCI and language resource levels, and showed that models implicitly personalise their answers towards the stereotypical culture of the prompt language throughout the layers.

Language bias in LLM evaluators

Oct 2025 – July 2026

- Demonstrated a language scoring bias in reward models and LLM-as-a-Judge (COLM 2026, ICBINB 2026).
- Showed that pairwise accuracy masks score miscalibration, and the bias correlates with language resource levels.
- Investigated why lower-resource languages score higher, a bias that correlates with model uncertainty.

Survey on explainability and interpretability of multilingual LLMs

Nov 2024 – Aug 2025

- Surveyed 200+ papers on multilingual explainability (EMNLP 2025).
- Categorised existing literature according to the explainability techniques employed, the multilingual tasks addressed, the languages investigated and available resources.
- Identified key challenges, distilled core findings and outlined promising avenues for future research.

Bilingual model merging and pretraining in small-capacity language models

Dec 2025 – present

- Evaluation of bilingual pretraining and model merging strategies in small-capacity language models (MME 2026).
- Systematic comparison of cross-lingual transfer and monolingual performance retention across training regimes.

Analysis of language-specific features of multilingual LLMs

Aug 2025 – present

- Definition, identification and ablation of language-specific features in multilingual LLMs (MRL 2025).
- Improvement of LLM low-resource language performance using language-specific features.

Improving NLP model explanations using human annotations

Dec 2021 – Mar 2024

- Developed a novel contrastive-inspired loss to incorporate human annotations into NLP classification in a model- and explainer-agnostic way (**NAACL Findings 2024**, LatinX in NLP, MSc thesis).
- Employed a multi-objective optimizer to explore the trade-off between the contrastive and the original losses.
- Significantly improved the plausibility of post-hoc explanations (relative increase of 3.49% for a language model) without substantially degrading model performance.

Design of a novel explainer for GNN node classification

July 2022 – Jan 2023

- *Distill n' Explain* (**AISTATS 2023**) first distills the original GNN into an interpretable one and then explains the latter.
- Designed and proved lemmas and theorems that guarantee the method's explanation faithfulness.
- The proposed explainer outperformed previous methods in explanation accuracy while being orders of magnitude faster.

Development of a visual analytics system to apply legal understandings

Feb 2022 – May 2025

- *LegalAnalytics* (AI & Law 2025) employs ML, NLP, XAI and visualization methods to assist judicial experts in the application of understandings from the Brazilian Supreme Court.
- Conducted a literature review on XAI in the legal domain (ASAIL 2025).

Application of machine learning to binding precedents

June 2021 – Apr 2025

- Explored ML, NLP and topological data analysis in legal documents that cite Brazilian binding precedents (AI & Law 2025).
- Managed the annotation of legal documents by experts (contribution to other papers).

Development of a visual analytics system to explore citations in legal documents

Aug 2020 – Feb 2022

- *LegalVis* (TVCG 2023, VIS 2022, BSc thesis) employs ML, NLP, XAI and data visualization to infer non-explicit citations in Brazilian legal documents.
- Tested a diverse set of NLP classifiers (Transformers, word embeddings and bag-of-words) and achieved high accuracy (96%) in identifying citations.
- Employed a model-agnostic explainer (LIME) to explain the inferred citations.

Study of legal language models with topological data analysis

Apr 2023 – Sept 2024

- Exploration of the intersection between NLP and topological data analysis in legal documents.
- Publications expected in 2025.

Study of training data attribution

Oct 2022 – Sept 2024

- Explored methods to attribute model predictions to training data.
- Investigated the intersection between attribution, datamodeling and machine unlearning.

Publications

Ej Zhou, **Lucas Resck**, Zheng Hui* and Anna Korhonen*. 2026. *Lower-Resource, Higher Scores*: Language Bias in LLM Evaluators. **COLM 2026**. *Equal advising.

Lucas Resck, Isabelle Augenstein and Anna Korhonen. 2026. Mitigating Cross-Lingual Cultural Inconsistencies in LLMs via Consensus-Driven Preference Optimisation. arXiv.

Ej Zhou, **Lucas Resck**, Zheng Hui and Anna Korhonen. 2026. Language-Dependent Miscalibration in Multilingual LLM Evaluators. Non-archival. I Can't Believe It's Not Better: Where Large Language Models Need to Improve (ICBINB 2026) at ICLR.

Suchir Salhan, Ej Zhou*, Laura Barbenel*, Aoife O'Driscoll*, Lily Goulder*, **Lucas Resck**, Catherine Arnett and Paula Buttery. 2026. *Glints of Gold or Troubling Waters?* Can a School of Merged Monolingual Goldfish Models Swim in Bilingual Seas? Non-archival. First Workshop on Multilingual Multicultural Evaluation (MME 2026) at EACL. *Equal contribution.

Lucas Resck, Isabelle Augenstein and Anna Korhonen. 2025. Explainability and Interpretability of Multilingual Large Language Models: A Survey. **EMNLP 2025**.

Lucas Resck* and Ej Zhou*. 2025. Dark Chest of Wonders: On the Specific Language-Specific Features of Multilingual Large Language Models. Non-archival extended abstract. 5th Workshop on Multilingual Representation Learning (MRL 2025) at EMNLP. *Equal contribution.

Lucas Resck, Felipe Moreno-Vera, Tobias Veiga, Gerardo Paucar, Ezequiel Fajreldines, Guilherme Klafke, Luis G. Nonato and Jorge Poco. 2025. LegalAnalytics: bridging visual explanations and workload streamline in Brazilian Supreme Court appeals. Artificial Intelligence and Law.

Lucas Resck, Felipe Moreno-Vera, Tobias Veiga, Gerardo Paucar, Ezequiel Fajreldines, Guilherme Klafke, Luis Gustavo Nonato and Jorge Poco. 2025. Explainable NLLP: Advancements in Explainable AI for Natural Legal Language Processing. Proceedings of the Seventh Workshop on Automated Semantic Analysis of Information in Legal Texts co-located with the 20th International Conference on Artificial Intelligence and Law (ICAAIL 2025). Short paper.

Raphaël Tinarrage, Henrique Ennes, **Lucas Resck**, Lucas T. Gomes, Jean R. Ponciano and Jorge Poco. 2025. Empirical analysis of binding precedent efficiency in Brazilian Supreme Court via case classification. Artificial Intelligence and Law.

Lucas Resck, Marcos M. Raimundo and Jorge Poco. 2024. Exploring the Trade-off Between Model Performance and Explanation Plausibility of Text Classifiers Using Human Rationales. **NAACL Findings 2024**. Also presented as a poster at the LatinX in NLP at NAACL 2024 workshop.

Lucas E. Resck, Jean R. Ponciano, Luis Gustavo Nonato and Jorge Poco. 2023. LegalVis: Exploring and Inferring Precedent Citations in Legal Documents. IEEE Transactions on Visualization and Computer Graphics (TVCG 2023). Presented at VIS 2022.

Tamara Pereira, Erik Nascimento, **Lucas E. Resck**, Diego Mesquita and Amauri Souza. 2023. Distill n' Explain: explaining graph neural networks using simple surrogates. **AISTATS 2023**.

Theses and Reports

Lucas Emanuel Resck Domingues. 2024. Balancing performance and explanation plausibility: a multi-objective approach to text classification with human rationales. MSc thesis, Getulio Vargas Foundation, Rio de Janeiro, Brazil.

Lucas Emanuel Resck Domingues. 2021. Inferring and Explaining Potential Citations to Binding Precedents in Brazilian Supreme Court Decisions. BSc thesis, Getulio Vargas Foundation, Rio de Janeiro, Brazil.

Lucas Emanuel Resck Domingues and Júlia Gandini Blahun. 2018. Circuits for Driving Low Power Direct Current Motors. Technical report, Federal Center for Technological Education of Minas Gerais, Varginha, Brazil.

Júlia Gandini Blahun, Luiza de Souza Pinto Regina and **Lucas Emanuel Resck Domingues**. 2016. Brazilian Robotics Olympiad – OBR'2016, Level II Practical Modality. Technical report, Federal Center for Technological Education of Minas Gerais, Varginha, Brazil.

Contribution to Other Papers

Beatriz Sabdin Chagas, Carla Marcondes Damian and Raphaël Tinarrage. 2022. The Impact of the Súmula Vinculante 26 on the Decrease of Similar Demands at the STF: a Quantitative Analysis With Machine Learning Models. XI International Meeting of CONPEDI. Law team: Beatriz Sabdin Chagas, Carla Marcondes Damian, Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira. Math team: Henrique Hennes, Jorge Poco, Jean Roberto Ponciano, **Lucas Resck**, Raphaël Tinarrage.

Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira and Raphaël Tinarrage. 2022. Regime Progression for Heinous Crimes in Brazilian Supreme Court (STF): an Empirical Analysis of Súmula Vinculante 26. XI International Meeting of CONPEDI. Law team: Beatriz Sabdin Chagas, Carla Marcondes Damian, Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira. Math team: Henrique Hennes, Jorge Poco, Jean Roberto Ponciano, **Lucas Resck**, Raphaël Tinarrage.

Teaching Experience

Professor Getulio Vargas Foundation (FGV) Professor of Introduction to Programming in Python in a Web Systems Development course.	<i>Rio de Janeiro, Brazil</i> <i>Jan 2023 – Apr 2023</i>
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Teaching Assistant Getulio Vargas Foundation (FGV) Teaching assistant of Ordinary Differential Equations, Calculus in Several Variables and Calculus in One Variable.	<i>Rio de Janeiro, Brazil</i> <i>Mar 2020 – June 2021</i>
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Work Experience

Consultant Interlink Capital Market Consulting Assisted the integration of NLP pipelines (from LLMs to classical models) in consulting projects.	<i>Remote</i> <i>Mar 2024 – Aug 2024</i>
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Summer Intern <i>EloGroup</i>	<i>Rio de Janeiro, Brazil</i> <i>Dec 2019 – Feb 2020</i>
Conducted time series analysis, exploratory data analysis, sanity checks on databases and data preprocessing.	
Summer Intern <i>PSR</i>	<i>Rio de Janeiro, Brazil</i> <i>Jan 2019 – Feb 2019</i>
Developed and implemented optimization models for maintenance schedules and dispatch of power plants, utilizing Julia and optimization packages.	

Honours, Awards and Scholarships

ELLIS PhD Student PhD student under the ELLIS PhD & Postdoc Program .	<i>Oct 2024 – Sept 2028</i>
Cambridge International Scholarship Cambridge Trust 's scholarship to cover tuition fees and maintenance costs.	<i>Oct 2024 – Sept 2027</i>
DDSA Visit Grant Danish Data Science Academy (DDSA) 's visit grant (15,000 DKK) for a research visit to the University of Copenhagen.	<i>Copenhagen, Denmark</i> <i>May 2026 – June 2026</i>
ELIAS Mobility Grant European Lighthouse of AI for Sustainability (ELIAS) 's mobility grant (€2,400) for a research visit to the University of Copenhagen.	<i>Copenhagen, Denmark</i> <i>May 2026 – June 2026</i>
Academic Development Fund Girton College's travel grant (£300) to partially cover travel expenses to EMNLP.	<i>Suzhou, China</i> <i>Oct 2025</i>
Young Researcher 11th Heidelberg Laureate Forum (HLF) Young Researcher.	<i>Sept 2024</i>
G-Research scholarship Registration and travel grant (€1,650) for EDS 2024 .	<i>Paris, France</i> <i>Aug 2024</i>
Google and Zendesk scholarship Registration and travel grant (€1,500) for LxMLS 2024 .	<i>Lisbon, Portugal</i> <i>July 2024</i>
NAACL Diversity and Inclusion Award Registration and partial travel grant (\$1,324) for NAACL 2024.	<i>Mexico City, Mexico</i> <i>June 2024</i>
MSc scholarship FGV scholarship: tuition fees and monthly stipend.	<i>Feb 2022 – May 2024</i>
Invited speaker Presentation <i>How the mathematics olympiads transformed my life</i> to motivate students from Colégio União, invited by Prof. Aguinaldo Borba.	<i>Três Corações, Brazil</i> <i>Mar 2023</i>
CAPES PROSUP scholarship Coordination for the Improvement of Higher Education Personnel (CAPES)'s Graduate Support Program for Private Education Institutions (PROSUP) scholarship to partially cover tuition fees.	<i>May 2022 – Feb 2023</i>
Academic distinguished undergraduate award Ranked 1st in my undergraduate class. Recognition of academic excellence (grades and research).	<i>Apr 2022</i>
Talent Selection Program Hosted by the Center for the Development of Mathematics and Sciences of FGV. - Scholarship holder (tuition fee and monthly stipend). - Selected based on my performance in mathematical olympiads and the entrance exam.	<i>Dec 2017 – Jan 2022</i>
PICME scholarship Undergraduate Research and Master's Program (PICME) scholarship. - It was only possible because of my mathematical olympiad medals before college. - I did research and took graduate courses during my undergraduate studies with a scholarship.	<i>Aug 2018 – July 2021</i>
High School Research Fellowship Scholarship holder at CEFET-MG and the National Council for Scientific and Technological Development (CNPq) in the High School Research Fellowship.	<i>Mar 2017 – Feb 2018</i>

1st place in the FGV entrance exam

Nov 2017

1st place in the entrance exam for the Applied Mathematics undergraduate program (24 candidates).

Olympiad medals

2012 – 2017

- Brazilian Public School Mathematics Olympiad (OBMEP): Gold (1 medal), silver (3) and bronze (1) medals and honourable mention (1).
- Brazilian Astronomy and Astronautics Olympiad (OBA): Silver (2) and bronze (1) medals.

Skills

Programming: Python, C++, Julia, R, MATLAB/Scilab, LaTeX

Machine Learning: PyTorch, scikit-learn, HuggingFace Transformers, TensorFlow, Keras

Technologies: Git, Pandas, NumPy, Linux

Languages: English (fluent, TOEFL: 112/120), Portuguese (native)

Other: First Aid for Sport (British Red Cross)

Service and Volunteering

[EACL 2027](#)

Athens, Greece

Website Chair.

Mar 2026 – present

- Developing, deploying and maintaining the open-source conference website infrastructure on GitHub Pages.
- Liaising with organising, program, workshop and tutorial chairs to centralise and publish calls, schedules and conference logistics.

Conference Reviewer

Nov 2025 – present

Reviewer for NeurIPS 2026, ICLR 2026 and VIS 2026.

[Language Technology Lab \(LTL\) Seminars](#)

Cambridge, UK

Co-organiser.

Oct 2025 – present

- Co-organisation of weekly research seminars, hosting speakers from top institutions (e.g., MIT, CMU, NYU; 8~10 international speakers per term).
- Management of logistics for hybrid events, welcoming attendees from the University and the wider public.

Cambridge University Brazilian Society (CUBS)

Cambridge, UK

President (Oct 2025 – present); Communications Officer (Dec 2024 – Oct 2025).

Dec 2024 – present

- Leading the committee and overseeing recruitment, financial structuring and digital infrastructure.
- Organising social events and managing communications to foster the Brazilian community in Cambridge.

EMNLP 2025

Suzhou, China

Volunteer at the [2025 Conference on Empirical Methods in Natural Language Processing](#).

Nov 2025

UAI 2025

Rio de Janeiro, Brazil

Volunteer at the [41st Conference on Uncertainty in Artificial Intelligence](#).

July 2025

FAccT 2024

Rio de Janeiro, Brazil

Volunteer at the [ACM Conference on Fairness, Accountability, and Transparency 2024](#).

June 2024

Academic Directory of Applied Mathematics at FGV

Aug 2018 – Aug 2019

Treasurer.

International Congress of Mathematicians (ICM) 2018

Rio de Janeiro, Brazil

Volunteer at the [ICM 2018](#).

Aug 2018

Events, Schools and Workshops Attended

[EMNLP 2025](#) and [MRL 2025 workshop](#)

Suzhou, China

Paper and extended abstract presentations and volunteer.

Nov 2025

[UAI 2025](#)

Rio de Janeiro, Brazil

Volunteer.

July 2025

<u>ASAIL 2025 workshop</u>	<i>Remote June 2025</i>
Paper presentation. Co-located with the 20th International Conference on Artificial Intelligence and Law (ICAAIL 2025).	
<u>11th Heidelberg Laureate Forum (HLF)</u>	<i>Heidelberg, Germany Sept 2024</i>
Selected as a Young Researcher and presented an artistic work at the Intercultural Science Art Project.	
<u>ELLIS Doctoral Symposium (EDS) 2024</u>	<i>Paris, France Aug 2024</i>
Presentation of a poster of the NAACL Findings 2024 paper.	
<u>Lisbon Machine Learning School (LxMLS) 2024</u>	<i>Lisbon, Portugal July 2024</i>
Presentation of a poster of the NAACL Findings 2024 paper.	
<u>NAACL 2024 and LatinX in NLP</u>	<i>Mexico City, Mexico June 2024</i>
Paper presentation.	
<u>FAccT 2024</u>	<i>Rio de Janeiro, Brazil June 2024</i>
Volunteer.	
<u>Tropical Probabilistic AI School 2024</u>	<i>Rio de Janeiro, Brazil Jan 2024</i>
Presentation of a poster of the NAACL Findings 2024 paper.	
Seminar for Postgraduate Students at FGV (SEPEMAp)	<i>Rio de Janeiro, Brazil Oct 2023</i>
Presentation of the NAACL Findings 2024 paper.	
<u>XLII Brazilian Congress of Applied and Computational Mathematics (CNMAC 2023)</u>	<i>Bonito, Brazil Sept 2023</i>
<u>Latin American Congress on Industrial and Applied Mathematics (LACIAM) 2023</u>	<i>Rio de Janeiro, Brazil Jan 2023</i>
<u>Summer School on Data Science 2023</u>	<i>Rio de Janeiro, Brazil Jan 2023</i>
<u>IEEE VIS: Visualization & Visual Analytics 2022</u>	<i>Oklahoma City, US Oct 2022</i>
Presentation of the TVCG 2023 paper.	
<u>8th Workshop on Mathematical Solutions for Industrial Problems</u>	<i>São Paulo, Brazil (virtual) Mar 2022</i>
Organised by the Research Center in Mathematics Applied to Industry (CeMEAI) at the University of São Paulo.	
<u>International Congress of Mathematicians (ICM) 2018</u>	<i>Rio de Janeiro, Brazil Aug 2018</i>
Volunteer.	
High School Research Course (<u>Mentores</u>)	<i>Virtual 2016</i>
Scholarship holder (CNPq) at the Plane Analytical Geometry course for medalists of OBMEP.	
High School Research Program (<u>PIC-Jr</u>)	<i>Varginha, Brazil 2013 – 2015</i>
Scholarship holder (CNPq) at the mathematics course for medalists of OBMEP.	